

Theory of Knot, Growth-Layer, and Fiber Modeling in the Board Generator

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This report describes the mathematical model used by the current board generator implementation. The presentation follows the code convention directly:

$$X = \text{width}, \quad Y = \text{thickness}, \quad Z = \text{length}.$$

The notation is therefore consistent with the updated growth-layer report and with the current implementation of the simulator. The theory is organized in the same order as the runtime pipeline: first the knot generator, then the growth-layer generator, and finally the fiber-orientation model.

1 Knot Generator

1.1 Knot Parameters

Each runtime knot n is described by the parameter vector

$$\kappa_n = (\theta_{0,n}, z_{0,n}, c_{1,n}, c_{2,n}, k_n, k_{p,n}, A_{\text{bump},n}, B_{\text{bump},n}, A_{\text{exp},n}, a_{1,n}, a_{2,n}, a_{3,n}, a_{4,n}, R_{L,n}, R_{D,n}). \quad (1)$$

The roles of these parameters are:

1. $\theta_{0,n}$ is the azimuth of the knot in the transverse (X, Y) plane.
2. $z_{0,n}$ is the longitudinal anchor position of the knot along the board.
3. $c_{1,n}$ and $c_{2,n}$ are the coefficients of the knot axis parabola.
4. k_n and $k_{p,n}$ control the growth-layer contraction and anisotropy.
5. $A_{\text{bump},n}$, $B_{\text{bump},n}$, and $A_{\text{exp},n}$ define the radial bump law.
6. $a_{1,n}, \dots, a_{4,n}$ are the coefficients of the quartic knot diameter law.
7. $R_{L,n}$ and $R_{D,n}$ are the live and dead transition radii.

The axis and the diameter law associated with one knot are written as

$$Z_{c,n}(\xi) = c_{1,n}\xi^2 + c_{2,n}\xi + z_{0,n}, \quad (2)$$

$$L_n(\xi) = a_{1,n}\xi^4 + a_{2,n}\xi^3 + a_{3,n}\xi^2 + a_{4,n}\xi. \quad (3)$$

The sequence model does not sample the full parameter vector directly. Instead it samples a dictionary row

$$d_m = (\theta_{0,m}, R_{L,m}, R_{D,m}, a_{1,m}, a_{2,m}, a_{3,m}, a_{4,m}, c_{1,m}, c_{2,m}), \quad (4)$$

and the remaining quantities are reconstructed during runtime decoding. In particular, $z_{0,n}$ is obtained from the longitudinal slot location, while k_n , $k_{p,n}$, $A_{\text{bump},n}$, $B_{\text{bump},n}$, and $A_{\text{exp},n}$ are derived from the sampled diameter profile.

1.2 Training Dataset

The knot-sequence training dataset is assembled from raw MATLAB files

knot_data.mat.

For each observed knot row, the preparation stage reads the matrices

Knot_param__Alfa_L100_Rlivelim_Rtot_Abump_Aexp,
Knot_param__Vertdiamparam,
Knot_param__Zcparam,

and constructs the runtime dictionary row d_m from the corresponding columns.

The raw longitudinal start coordinate is read as $z_{0,m}$ and aligned to a regular slot lattice of spacing Δz :

$$z_{0,m}^{(\Delta)} = \text{round}\left(\frac{z_{0,m}}{\Delta z}\right) \Delta z. \quad (5)$$

If two consecutive knots land in the same slot after rounding, the later one is shifted forward by one slot. Negative slots are discarded.

Each log then becomes a discrete sequence

$$s^{(\ell)} = (s_1^{(\ell)}, s_2^{(\ell)}, \dots, s_{T_\ell}^{(\ell)}), \quad s_j^{(\ell)} \in \{0, 1, \dots, M\}, \quad (6)$$

where 0 denotes an empty slot and every nonzero symbol refers to one knot dictionary entry. The sequence is segmented into overlapping windows of length $N_s = 101$ with overlap 97, so the stride is four slots. For each window

$$w = (t_1, t_2, \dots, t_{101}), \quad (7)$$

the supervised training pair is

$$\text{input} = (t_1, \dots, t_{100}), \quad \text{target} = (t_2, \dots, t_{101}). \quad (8)$$

In addition to the runtime dictionary, the data-preparation code stores an auxiliary 23-dimensional shape descriptor. The first three entries are (θ_0, R_L, R_D) ; the next ten entries are samples of the diameter law along normalized axial positions; and the final ten entries are samples of the axis law along normalized positions. This auxiliary representation is retained for analysis and compatibility. The actual sequence model still operates on the tokenized knot sequences.

1.3 Clustering of Knots in the Training Set

Two tokenization modes are supported.

In the direct mode, each nonzero token corresponds to exactly one dictionary row. In the clustered mode, the dictionary rows are first normalized columnwise to $[0, 1]$:

$$\hat{d}_m = \frac{d_m - d_{\min}}{d_{\max} - d_{\min}}, \quad (9)$$

and the nonzero rows are grouped into Q clusters by k -means. The clustering problem is

$$\min_{\{\mu_q\}, \{c_m\}} \sum_{m=1}^M \left\| \hat{d}_m - \mu_{c_m} \right\|_2^2, \quad c_m \in \{1, \dots, Q\}, \quad (10)$$

where μ_q is the centroid of cluster q and c_m is the assigned cluster index of dictionary row m .

Token 0 is always reserved for the no-knot state. Therefore the learned vocabulary is

$$\mathcal{V} = \{0, 1, \dots, Q\} \quad (11)$$

in clustered mode and

$$\mathcal{V} = \{0, 1, \dots, M\} \quad (12)$$

in direct mode.

The clustering stage stores both a forward map and an inverse membership map. During runtime sampling, if a clustered token $q > 0$ is drawn, one concrete dictionary row is chosen uniformly from the members of that cluster. This preserves stochastic diversity while keeping the language-model vocabulary compact.

1.4 LSTM Sequence Model

Let $t_u \in \mathcal{V}$ denote the token at slot u . The knot generator models the autoregressive sequence distribution

$$P(t_1, \dots, t_T) = \prod_{u=1}^T P(t_u \mid t_1, \dots, t_{u-1}). \quad (13)$$

Each token is embedded into a continuous vector

$$e_u = E[t_u], \quad (14)$$

where the embedding matrix E is initialized from the normalized dictionary rows or cluster centroids. The embedded sequence is processed by an LSTM,

$$(h_u, c_u) = \text{LSTM}(e_u, h_{u-1}, c_{u-1}), \quad (15)$$

followed by a linear prediction head

$$\ell_u = Wh_u + b, \quad P(t_{u+1} = q \mid t_{\leq u}) = \text{softmax}(\ell_u)_q. \quad (16)$$

Training minimizes a weighted next-token cross-entropy:

$$\mathcal{L} = - \sum_{u=1}^{T-1} w_{t_{u+1}} \log P(t_{u+1} \mid t_{\leq u}), \quad (17)$$

where the no-knot token 0 receives a reduced class weight. This is necessary because empty slots dominate the longitudinal sequence distribution and would otherwise overwhelm the nonzero knot tokens.

At inference time, the generator samples a fresh sequence of length matching the board slot count. The sampler supports both top- k truncation and nucleus filtering. If a trained checkpoint is unavailable, the runtime can fall back to a first-order Markov chain estimated from empirical token transitions in the training dataset.

1.5 Runtime Decoding and Post-Processing

After a token sequence has been sampled, token i is assigned to the longitudinal slot

$$z_i = z_{\min} + i\Delta z, \quad i = 1, \dots, N_{\text{slot}}. \quad (18)$$

Zero tokens are discarded. In clustered mode, every surviving token is decoded to a concrete dictionary row as described above. Bad dictionary rows are removed, and the quartic diameter law is summarized by

$$L_{100} = a_1 100^4 + a_2 100^3 + a_3 100^2 + a_4 100. \quad (19)$$

Only knots with

$$L_{100,\min} \leq L_{100} \leq L_{100,\max} \quad (20)$$

are kept.

The simulator then derives the bump-law parameters through

$$B_{\text{bump}} = 2, \quad (21)$$

$$A_{\text{bump}} = |0.0217 L_{100} - 0.2|, \quad (22)$$

$$A_{\text{exp}} = 0.0056 L_{100} + 1.96, \quad (23)$$

and sets

$$k = 0.99, \quad k_p = 0.95. \quad (24)$$

For generated knots, the dead-zone width is repaired when necessary so that

$$R_D - R_L \geq \Delta_{\min}. \quad (25)$$

Finally, the candidate knot set is checked for pairwise overlap by sampling each knot axis together with its local diameter envelope. Two samples from different knots are deemed overlapping when

$$\|p_i - p_j\|_2 < r_i + r_j - \delta, \quad (26)$$

where p_i and p_j are sampled axis points, r_i and r_j are the associated local radii, and δ is the configured clearance. Sequences with overlapping knots are rejected and resampled until either a valid candidate is found or the retry budget is exhausted.

2 Growth-Layer Generator

2.1 Base Mesh and Nominal Growth Layers

The simulator starts from a Cartesian grid

$$\mathcal{G} = \{(X_{ijk}, Y_{ijk}, Z_{ijk})\}, \quad (27)$$

constructed from one-dimensional coordinate arrays in the board domain. On this grid the transverse polar coordinates are

$$\Theta = \text{atan2}(Y, X), \quad R = \sqrt{X^2 + Y^2}. \quad (28)$$

For each growth-layer index i , the model stores a spline $s_i(\Theta)$ that defines the nominal layer radius:

$$R_{o,i}(\Theta) = s_i(\Theta). \quad (29)$$

Without knots, crook, or taper, the implicit field of the i -th layer would be

$$g_i^{\text{nom}}(X, Y, Z) = X^2 + Y^2 - R_{o,i}(\Theta)^2. \quad (30)$$

The zero level set of this field is the nominal growth-layer surface.

2.2 Crook and Taper

The growth-layer generator does not depend on how crook and taper were sampled. It only assumes that the current board realization provides two transverse displacement functions,

$$c_x(Z), \quad c_y(Z),$$

and one taper function,

$$t(Z).$$

These functions may come from a random multi-component crook model, from manual parameters, or from legacy quadratic terms. The growth-layer equations consume them only through

$$X_c = X + c_x(Z), \quad Y_c = Y + c_y(Z), \quad (31)$$

$$\tilde{R}_{o,i}(\Theta, Z) = R_{o,i}(\Theta) - t(Z). \quad (32)$$

Thus crook translates the transverse coordinates as a function of the longitudinal position, whereas taper reduces the local effective ring radius.

2.3 Rotation into the Knot Frame

For a fixed knot n , the crook-corrected transverse point can be rotated directly in Cartesian form. The local knot-frame coordinates are

$$\xi_n = X_c \cos \theta_{0,n} - Y_c \sin \theta_{0,n}, \quad \eta_n = X_c \sin \theta_{0,n} + Y_c \cos \theta_{0,n}. \quad (33)$$

The coordinate ξ_n is the radial coordinate in the rotated transverse plane, while η_n is the orthogonal tangential coordinate. This expression is exactly equivalent to first writing $X_c = r \cos \phi$, $Y_c = r \sin \phi$ and then evaluating

$$\xi_n = r \cos(\phi + \theta_{0,n}), \quad \eta_n = r \sin(\phi + \theta_{0,n}),$$

but it avoids an unnecessary conversion to polar coordinates.

2.4 Knot Axis and Influence Metric

The knot centerline is modeled by a parabola in the (ξ_n, Z) plane:

$$Z_{c,n}(\xi_n) = c_{1,n}\xi_n^2 + c_{2,n}\xi_n + z_{0,n}. \quad (34)$$

The signed longitudinal offset from the knot axis is

$$\Delta Z_n = Z - Z_{c,n}(\xi_n). \quad (35)$$

The anisotropic knot-deviation metric is then

$$p_n = \sqrt{\Delta Z_n^2 + \frac{1}{k_{p,n}^2} (\xi_n^2 + \eta_n^2) \text{atan2}(\eta_n, \xi_n)^2}. \quad (36)$$

To avoid singular behavior near the knot axis, the implementation applies a smooth lower clamp:

$$\bar{p}_n = \sigma(\alpha(p_n - p_{\min})) p_n + [1 - \sigma(\alpha(p_n - p_{\min}))] p_{\min}, \quad (37)$$

where $\sigma(s) = (1 + e^{-s})^{-1}$ is the logistic sigmoid. The maximum active influence distance is

$$p_{\max,n} = \left(\frac{A_{\text{bump},n} \tilde{R}_{o,i}^{A_{\text{exp},n}-1}}{1 - k_n} \right)^{1/B_{\text{bump},n}}. \quad (38)$$

2.5 Living-Knot Radius Law

When the dead-knot extension is disabled, the knot affects the layer only in the forward half-space $\xi_n \geq 0$ and only while $\bar{p}_n \leq p_{\max,n}$. The effective layer radius for knot n is

$$R_{i,n} = \begin{cases} \tilde{R}_{o,i}, & \bar{p}_n > p_{\max,n} \text{ or } \xi_n < 0, \\ k_n \tilde{R}_{o,i} + A_{\text{bump},n} \tilde{R}_{o,i}^{A_{\text{exp},n}} \bar{p}_n^{-B_{\text{bump},n}}, & \text{otherwise.} \end{cases} \quad (39)$$

If knot deviations are disabled entirely, then the simulator simply sets

$$R_{i,n} = \tilde{R}_{o,i}. \quad (40)$$

2.6 Dead-Knot Transition

When dead-knot logic is enabled, the model introduces a transition annulus

$$R_{L,n} \leq \tilde{R}_{o,i} \leq R_{D,n}. \quad (41)$$

The code defines

$$S_n = k_n R_{L,n}, \quad T_n = A_{\text{bump},n} R_{L,n}^{A_{\text{exp},n}}, \quad K_{\text{red},n} = \frac{\tilde{R}_{o,i} - R_{L,n}}{R_{D,n} - R_{L,n}}, \quad (42)$$

$$\Delta R_n = R_{D,n} - R_{L,n}, \quad \gamma = \frac{1}{1.5}, \quad (43)$$

$$\Delta p_n = \text{Re} \left(\sqrt{\Delta R_n^2 - \gamma^2 (R_{D,n} - \tilde{R}_{o,i})^2} \right) - \sqrt{1 - \gamma^2} \Delta R_n, \quad (44)$$

$$\bar{p}_{n,\text{alt}} = \bar{p}_n + \Delta p_n. \quad (45)$$

Within the transition region the effective radius becomes

$$R_{i,n}^{\text{dead}} = K_{\text{red},n} R_{D,n} + (1 - K_{\text{red},n}) \left(S_n + T_n \bar{p}_{n,\text{alt}}^{-B_{\text{bump},n}} \right). \quad (46)$$

The full dead-knot piecewise law is therefore

$$R_{i,n} = \begin{cases} \tilde{R}_{o,i}, & \bar{p}_n > p_{\text{max},n} \text{ or } \xi_n < 0 \text{ or } \tilde{R}_{o,i} > R_{D,n}, \\ R_{i,n}^{\text{dead}}, & R_{L,n} \leq \tilde{R}_{o,i} \leq R_{D,n}, \\ k_n \tilde{R}_{o,i} + A_{\text{bump},n} \tilde{R}_{o,i}^{A_{\text{exp},n}} \bar{p}_n^{-B_{\text{bump},n}}, & \text{otherwise.} \end{cases} \quad (47)$$

Finally the implementation enforces

$$R_{i,n} \leftarrow \max(R_{i,n}, 1). \quad (48)$$

2.7 Growth-Layer Field and Combination of Knots

For one layer i and one knot n , the knot-specific implicit field is

$$g_{i,n}(X, Y, Z) = \xi_n^2 + \eta_n^2 - R_{i,n}^2. \quad (49)$$

The actual growth-layer field is the lower envelope over all knots:

$$g_i(X, Y, Z) = \min_n g_{i,n}(X, Y, Z). \quad (50)$$

The deformed growth-layer surface is the zero isosurface

$$\Gamma_i = \{(X, Y, Z) : g_i(X, Y, Z) = 0\}. \quad (51)$$

If no knots are present, the simulator still produces a valid ring field:

$$g_i(X, Y, Z) = X_c^2 + Y_c^2 - \max(\tilde{R}_{o,i}, 1)^2. \quad (52)$$

When dead-knot mode is active, the implementation applies an additional nesting correction from the outside inward:

$$\tilde{g}_N = g_N, \quad \tilde{g}_i = \max(g_i, \tilde{g}_{i+1}), \quad i = N - 1, \dots, 1. \quad (53)$$

This prevents an inner layer from protruding outside the next outer layer in the dead-knot transition zone.

2.8 Knot Scalar Field

Independently of the ring field, the simulator constructs a knot-body field. The quartic diameter law $L_n(\xi_n)$ is first evaluated. In dead-knot mode, the law is tapered after $R_{L,n}$ by freezing its value at $R_{L,n}$ and multiplying it by a quadratic decay

$$h(t) = 1 - t^2, \quad t = \frac{\xi_n - R_{L,n}}{R_{D,n} - R_{L,n}}, \quad (54)$$

with $L_n(\xi_n) = 0$ for $\xi_n > R_{D,n}$.

The knot scalar field is

$$K_n(X, Y, Z) = \Delta Z_n^2 + \frac{\eta_n^2}{k_{p,n}^2} - \left(\frac{L_n(\xi_n)}{2} \right)^2. \quad (55)$$

The combined field is

$$K_{\min}(X, Y, Z) = \min_n K_n(X, Y, Z). \quad (56)$$

By construction, $K_n < 0$ indicates the interior of the knot body, $K_n = 0$ is the nominal knot boundary, and $K_n > 0$ is outside the knot.

For visualization and masking, the code also keeps separate live and dead subsets:

$$K_{n,\text{live}} = \begin{cases} K_n, & \xi_n \leq R_{L,n}, \\ \text{NaN}, & \text{otherwise}, \end{cases} \quad (57)$$

$$K_{n,\text{dead}} = \begin{cases} K_n, & R_{L,n} < \xi_n \leq R_{D,n}, \\ \text{NaN}, & \text{otherwise}. \end{cases} \quad (58)$$

2.9 Exported World-Space Knot Axes

The implementation also exports explicit world-space knot axes. For a sampled radial coordinate ξ with $\eta = 0$, the axis centerline is

$$Z_n^{\text{axis}}(\xi) = c_{1,n}\xi^2 + c_{2,n}\xi + z_{0,n}, \quad (59)$$

$$X_n^{\text{axis}}(\xi) = \xi \cos \theta_{0,n}, \quad Y_n^{\text{axis}}(\xi) = -\xi \sin \theta_{0,n}. \quad (60)$$

Crook compensation then maps the axis into world coordinates:

$$X_n^{\text{axis,world}}(\xi) = X_n^{\text{axis}}(\xi) - c_x(Z_n^{\text{axis}}(\xi)), \quad (61)$$

$$Y_n^{\text{axis,world}}(\xi) = Y_n^{\text{axis}}(\xi) - c_y(Z_n^{\text{axis}}(\xi)), \quad (62)$$

$$Z_n^{\text{axis,world}}(\xi) = Z_n^{\text{axis}}(\xi). \quad (63)$$

These are the quantities exported to the viewer and to downstream geometry consumers.

3 Fiber Orientation Model

3.1 Local Potential-Flow Model in the Knot Frame

The fiber-orientation model is solved after the growth-layer field has been constructed. For each knot, the local disturbance is modeled in the tangential-longitudinal plane of the knot frame. The effective obstacle width is

$$b_n = k_{p,n} L_n(\xi_n), \quad (64)$$

and the code solves for two flow parameters, a_n and G_n .

The default approximation used to initialize the solver is

$$a_n^{\text{approx}} = \frac{\sqrt{\nu} L_n(\xi_n) (b_n/2)}{\sqrt{(2\nu + 1)L_n(\xi_n)^2 + (4\nu - 4)(b_n/2)^2}}, \quad \nu = 7. \quad (65)$$

If the exact mode is enabled, this initial guess is refined by Newton iterations applied to the same residual system used in the MATLAB implementation. Once a_n has been determined, the source strength is

$$G_n = \frac{\pi L_n(\xi_n)^2 - 4\pi a_n^2}{4a_n}. \quad (66)$$

3.2 Stream-Function Derivatives and Local Fiber Direction

Let $c_n(\eta_n, Z)$ denote the local stream function. Its derivatives, as evaluated by the implementation, are

$$\frac{\partial c_n}{\partial \eta_n} = \frac{G_n}{2\pi} \left[\frac{a_n + \Delta Z_n}{\eta_n^2 + (a_n + \Delta Z_n)^2} + \frac{a_n - \Delta Z_n}{\eta_n^2 + (a_n - \Delta Z_n)^2} \right] + 1, \quad (67)$$

$$\frac{\partial c_n}{\partial Z} = -\frac{G_n}{2\pi} \left[\frac{\eta_n}{\eta_n^2 + (a_n + \Delta Z_n)^2} - \frac{\eta_n}{\eta_n^2 + (a_n - \Delta Z_n)^2} \right]. \quad (68)$$

The local fiber direction is tangent to the streamlines, and therefore the unnormalized tangential and longitudinal components are

$$u_{\eta,n} = -\frac{\partial c_n}{\partial Z}, \quad u_{Z,n} = \frac{\partial c_n}{\partial \eta_n}. \quad (69)$$

After normalization,

$$\hat{u}_{\eta,n} = \frac{u_{\eta,n}}{\sqrt{u_{\eta,n}^2 + u_{Z,n}^2}}, \quad \hat{u}_{Z,n} = \frac{u_{Z,n}}{\sqrt{u_{\eta,n}^2 + u_{Z,n}^2}}. \quad (70)$$

This local field lives in the (η_n, Z) plane. In other words, the potential-flow model provides the tangential and longitudinal components directly; the radial component is recovered later by enforcing tangency to the growth-layer surface.

3.3 Rotation to Board Coordinates and Crook Correction

The tangential basis vector of the knot frame maps into the board coordinates as

$$\mathbf{e}_{\eta,n} = (\sin \theta_{0,n}, \cos \theta_{0,n}, 0),$$

and the longitudinal basis vector is the board-length direction

$$\mathbf{e}_Z = (0, 0, 1).$$

Hence the local flow field becomes

$$\mathbf{t}_n^{(0)} = \hat{u}_{\eta,n} \mathbf{e}_{\eta,n} + \hat{u}_{Z,n} \mathbf{e}_Z, \quad (71)$$

or componentwise

$$t_{x,n}^{(0)} = \hat{u}_{\eta,n} \sin \theta_{0,n}, \quad t_{y,n}^{(0)} = \hat{u}_{\eta,n} \cos \theta_{0,n}, \quad t_{z,n}^{(0)} = \hat{u}_{Z,n}. \quad (72)$$

If crook-fiber correction is enabled, the model subtracts the local pith tangent from the transverse components. Writing

$$c'_x(Z) \approx \frac{c_x(Z + \delta) - c_x(Z - \delta)}{2\delta}, \quad c'_y(Z) \approx \frac{c_y(Z + \delta) - c_y(Z - \delta)}{2\delta}, \quad (73)$$

with $\delta = 0.1$ mm, the corrected field is

$$t_{x,n}^{(1)} = t_{x,n}^{(0)} - c'_x(Z), \quad t_{y,n}^{(1)} = t_{y,n}^{(0)} - c'_y(Z), \quad t_{z,n}^{(1)} = t_{z,n}^{(0)}. \quad (74)$$

3.4 Tangency to the Growth Layers

The clear-wood fibers must remain tangent to the nearest growth-layer surface. Let

$$i^*(X, Y, Z) = \underset{i}{\operatorname{argmin}} |g_i(X, Y, Z)| \quad (75)$$

be the index of the nearest layer and let

$$\mathbf{n} = \frac{\nabla g_{i^*}}{\|\nabla g_{i^*}\|_2} = (n_x, n_y, n_z) \quad (76)$$

be its normalized normal vector.

The local potential-flow field $\mathbf{t}_n^{(1)}$ has no explicit radial component. The solver therefore adds a radial correction of the form

$$\delta_n \mathbf{e}_{\xi, n}, \quad \mathbf{e}_{\xi, n} = (\cos \theta_{0, n}, -\sin \theta_{0, n}, 0), \quad (77)$$

and chooses δ_n such that the corrected vector is tangent to the layer:

$$\mathbf{n} \cdot (\mathbf{t}_n^{(1)} + \delta_n \mathbf{e}_{\xi, n}) = 0. \quad (78)$$

Solving for δ_n gives

$$\delta_n = -\frac{n_x t_{x, n}^{(1)} + n_y t_{y, n}^{(1)} + n_z t_{z, n}^{(1)}}{n_x \cos \theta_{0, n} - n_y \sin \theta_{0, n} + \varepsilon}, \quad (79)$$

with a small numerical stabilizer $\varepsilon > 0$. The final tangent-consistent vector prior to knot selection is

$$\mathbf{t}_n = \mathbf{t}_n^{(1)} + \delta_n \mathbf{e}_{\xi, n}, \quad (80)$$

followed by normalization.

3.5 Multi-Knot Competition and In-Knot Override

If several knots contribute candidate fiber vectors at the same point, the implementation selects the one with the strongest longitudinal effect:

$$n^* = \underset{n}{\operatorname{argmax}} |t_{z, n}|. \quad (81)$$

Outside invalid regions, the fiber field is therefore

$$\mathbf{f}_{\text{clear}}(X, Y, Z) = \mathbf{t}_{n^*}(X, Y, Z). \quad (82)$$

If the computation produces undefined values, the field defaults to straight clear wood,

$$\mathbf{f}_{\text{default}} = (0, 0, 1). \quad (83)$$

Inside the knot body, the simulator may replace the clear-wood field by the tangent to the knot axis itself. Because the axis law is $Z_{c, n}(\xi_n)$, the knot-axis tangent in the local knot frame is

$$\boldsymbol{\tau}_n^{\text{kf}} = \frac{(1, 0, 2c_{1, n}\xi_n + c_{2, n})}{\sqrt{1 + (2c_{1, n}\xi_n + c_{2, n})^2}}. \quad (84)$$

Mapping this vector into board coordinates gives

$$\boldsymbol{\tau}_n = \left(\frac{\cos \theta_{0, n}}{\sqrt{1 + (2c_{1, n}\xi_n + c_{2, n})^2}}, -\frac{\sin \theta_{0, n}}{\sqrt{1 + (2c_{1, n}\xi_n + c_{2, n})^2}}, \frac{2c_{1, n}\xi_n + c_{2, n}}{\sqrt{1 + (2c_{1, n}\xi_n + c_{2, n})^2}} \right). \quad (85)$$

The override is applied wherever

$$K_n(X, Y, Z) < K_{\text{in}}, \quad (86)$$

with K_{in} given by the configured interior threshold. Two optional refinements are supported:

1. The override can be suppressed in the dead-knot region by ignoring points where $K_{n,\text{dead}}$ is finite.
2. The override direction can be sign-reversed for $\Delta Z_n > 0$ so that the axis tangent flips on the positive longitudinal side of the knot.

The resulting fiber field can therefore be summarized as

$$\mathbf{f}(X, Y, Z) = \begin{cases} \boldsymbol{\tau}_n, & \text{inside the active knot-override region,} \\ \mathbf{t}_{n*}, & \text{otherwise, when a valid knot-induced field exists,} \\ (0, 0, 1), & \text{in the remaining clear-wood default region.} \end{cases} \quad (87)$$

3.6 Interpretation

The final fiber-orientation model combines three mechanisms.

First, the potential-flow construction bends fibers around the knot in the local tangential-longitudinal plane. Second, the tangency correction injects the radial component required for the fibers to remain tangent to the nearest growth layer. Third, the optional in-knot override replaces the clear-wood field by the knot-axis tangent inside the knot body. This yields a model that is consistent with the growth-layer geometry, respects the updated coordinate convention, and remains close to the theory used in the MATLAB implementation.